



Interference analysis and workspace optimization of 3-RRR spherical parallel mechanism

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ABSTRACT

The causes of restriction on the working scope of spherical parallel mechanism (SPM) were analyzed. A systematic design methodology (Least Interference Design Methodology, LIDeM) for reducing interference among links and increasing workspace of mechanism was also developed in this study. Focusing on 3-RRR SPM, the optimal structure was given according to the LIDeM theory. The optimum values of the modified link ratio β_1 and β_2 were obtained through numerical simulation of its mathematical model. In addition, the total workspace mapping, singularity loci and interference regions of the optimum 3-RRR SPM were also attained in this study by computational simulation. This work improves the total workspace of 3-RRR SPM and could be applied on some other revolute-joint mechanisms.

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1. Introduction

In many fields such as the robot industry, aerospace engineering, medical care industry, and illuminating system, some mechanisms which can implement twirling motion are widely used, especially in some cases such as industrial manipulator, telescope orienting device, surgical endoscope, etc. Those applications require the mechanical part to have a wide working range, free-collision motions, and the ability to be located rapidly at arbitrary direction within the workspace. Furthermore, this kind of device is also becoming a part of human's modern life style. It is likely that this kind of orienting mechanisms or robots will be seen at home in the near future. Nowadays, the research on spatial mechanism has turned into one of the hottest issues around the world.

The most typical mechanism to orient in a 3D space is spherical mechanism. Numerous researches are focusing on one or two degrees of freedom (DOF) mechanisms, e.g., 4-bar and 5-bar spherical mechanism [1,2]. Whereas these mechanisms' workspaces are restricted by their DOFs, they are not sufficed in some situations where high dexterity is needed. However, there is another spherical mechanism called Spherical Parallel Mechanism (SPM) which could be a good option. There are many advantages that make 3DOF SPM irreplaceable, such as wider workspace, lower moments of inertia, compact structure and high stiffness. Only with the prismatic joints and revolute joints, seven arrangements of 3DOF SPM limbs can be obtained: 3-RRR, 3-RRP, 3-RPR, 3-PRR, 3-RPP, 3-PRP and 3-PPR [3]. Many researchers, such as Enferadi and Tootoonchi [4], have been dedicating their work on these mechanisms. They raised an idea to replace 3-RRP SPM by the 3-PRP one which is based on the forward kinematics analysis and the isotropic structure design. Bai et al. [5] investigated the forward kinematic problem of 3-RRR SPM by dividing it into two 4-bar linkages, which avoids solving unnecessary nonlinear equations and leads to less calculations. As the most famous 3-RRR SPM, the Agile Eye [6] shows its superior performance and can be used as a robot wrist or a monitoring device, and in particular its

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wider orienting range compares to Delta robot [7] which has already been used in surgical operation. Although the Agile Eye can point any orientation within a cone of approximately 140° and $\pm 30^\circ$ in torsion [8], there is still the limitation that its working scope cannot cover a whole sphere. Chablat and Wenger's [9] utilized 3-RRR SPM as the vertebra of an eel robot. For increasing the workspace, they raised some solutions for avoiding the interference between linkages and motors. However, those solutions cannot settle general problems.

Most of those researches on SPM kinematics ignore the influence of interference among the mechanical components, which causes the shrinking of total workspace. In order to attain the optimal workspace of SPM, investigations on avoiding interference among components need to be done. Ling and Chas [10] addressed an apriori design method of preventing interference in planar mechanism. The main idea is to put the interfered components onto different planar surfaces. Based on their study, Sen and Joshi [11] made some improvements on the multi-layer approach by changing the shape of links and joints and providing alternative schemes through matrices. When the radius of curvature increases to infinity, planar mechanism can be regarded as a particular case of spherical mechanism. Accordingly, these approaches on planar mechanism can be employed for spherical mechanism. Through tracing a reference point, Merlet et al. [12] came up with the determination of workspace boundary of planar parallel mechanism.

It can be seen that most of the analyses and syntheses of SPM hardly take into account the influence by mechanical interference. Although there are some approaches dealing with that topic in planar mechanism, they cannot provide an optimal solution either. All these studies need complements before the application onto SPM. Consequently, the study aims to develop a design methodology for SPM optimal workspace and least interference. Based on that, the optimum structure and design parameters of 3-RRR SPM are derived.

2. Least Interference Design Methodology (LiDeM)

2.1. Position analysis of 3-RRR SPM

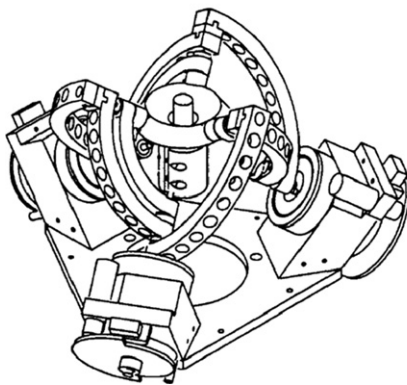
In order to investigate the interference effecting on the workspace of 3-RRR SPM, the study takes the Agile Eye as an instance (Fig. 1a). After the design methodology is applied, the increment of its workspace can be seen evidently. The positions of each joint can be represented by a vector.

In Fig. 1b, U_1 , U_2 and U_3 imply three upper limbs, and L_1 , L_2 and L_3 imply three lower ones; $\mathbf{m}_1$, $\mathbf{m}_2$ and $\mathbf{m}_3$ are along the axial directions of motor shafts, and coincide with the coordinate axes $\mathbf{x}$, $\mathbf{y}$ and $\mathbf{z}$ respectively:

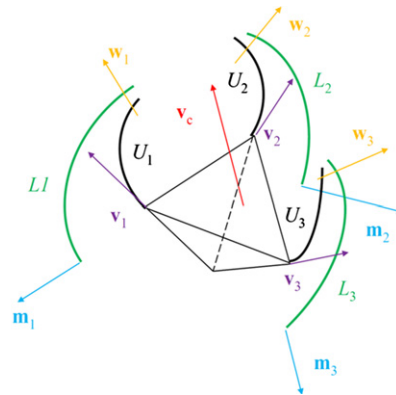
$$\begin{cases} \mathbf{m}_1 = \mathbf{i} + 0\mathbf{j} + 0\mathbf{k} \\ \mathbf{m}_2 = 0\mathbf{i} + \mathbf{j} + 0\mathbf{k} \\ \mathbf{m}_3 = 0\mathbf{i} + 0\mathbf{j} + \mathbf{k} \end{cases} \quad (1)$$

$\mathbf{w}_1$, $\mathbf{w}_2$ and $\mathbf{w}_3$ represent the revolute joints between upper limbs and lower limbs:

$$\begin{cases} \mathbf{w}_1 = 0\mathbf{i} + \cos\theta_1\mathbf{j} + \sin\theta_1\mathbf{k} \\ \mathbf{w}_2 = \sin\theta_2\mathbf{i} + 0\mathbf{j} + \cos\theta_2\mathbf{k} \\ \mathbf{w}_3 = \cos\theta_3\mathbf{i} + \sin\theta_3\mathbf{j} + 0\mathbf{k} \end{cases} \quad (2)$$



a) 3D drawing of the Agile Eye [6]



b) Kinematic diagram of the Agile Eye

Fig. 1. The Agile Eye by University of Laval, a. 3D drawing of the Agile Eye [6], b. Kinematic diagram of the Agile Eye.

where θ_n ($n = 1, 2, 3$) is the angular displacement of each input motor. Since the terminal point of unit vector $\mathbf{v}_c$ is on a spherical surface, the orienting vector of the end-effector can be represented as below:

$$\mathbf{v}_c = \cos\phi_1 \sin\phi_2 \mathbf{i} + \sin\phi_1 \sin\phi_2 \mathbf{j} + \cos\phi_2 \mathbf{k} \quad (3)$$

where ϕ_1 and ϕ_2 are the tilt angles [13].

$\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_3$ are defined as the joints between the end-effector and upper limbs, and are orthogonal to each other. The relation between $\mathbf{v}_n$ and $\mathbf{v}_c$ can be seen from the equilibrium:

$$\mathbf{v}_n \cdot \mathbf{v}_c = \frac{\sqrt{3}}{3}, \text{ for } n = 1, 2, 3. \quad (4)$$

Combine Eqs. (3) and (4), and $\mathbf{v}_n$ can be attained by the product of Euler Quaternion (an extension of complex numbers):

$$\mathbf{v}_n = \mathbf{q}_c \mathbf{q}_e \mathbf{q}_{e0} \mathbf{m}_n \mathbf{q}_{e0}^{-1} \mathbf{q}_e^{-1} \mathbf{q}_c^{-1} \quad (5)$$

where $\mathbf{q}_{e0} = \cos \rho + \mathbf{e}_0 \sin \rho = \exp(\mathbf{e}_0 \rho)$ is the complex exponential representation for the rotation around the axis $\mathbf{e}_0 = \cos 45^\circ \mathbf{i} - \sin 45^\circ \mathbf{j} + 0 \mathbf{k}$ by $\rho = \arccos \sqrt{3}/3$. Similarly, $\mathbf{q}_e = \exp(\mathbf{e}_r \phi_2/2)$ defines the rotation around the axis $\mathbf{e}_r = \cos \phi_1 \mathbf{i} + \sin \phi_1 \mathbf{j} + 0 \mathbf{k}$ by ϕ_2 , and $\mathbf{q}_c = \exp(\mathbf{v}_c \omega/2)$ denotes the rotation around the axis $\mathbf{v}_c$ by the torsion angle ω [13]; $\mathbf{q}_{e0}^{-1}$, $\mathbf{q}_e^{-1}$ and $\mathbf{q}_c^{-1}$ are the conjugates of $\mathbf{q}_{e0}$, $\mathbf{q}_e$ and $\mathbf{q}_c$ respectively.

Hence, the forward kinematic problem can be solved from the equation:

$$\mathbf{w}_n = \mathbf{v}_n \times \mathbf{m}_n. \quad (6)$$

Accordingly, $\mathbf{r}_{Un}$ ($\mathbf{r}_{Ln}$) is the collection of all the position vectors of the points on the upper (lower) limb of the mechanism, and can be acquired through Quaternion operation:

$$\mathbf{r}_{Un} = e^{\mathbf{e}_n^U \alpha_1} \mathbf{v}_n, \alpha_1 \in [0, \frac{\pi}{2}] \quad (7)$$

$$\mathbf{r}_{Ln} = e^{\mathbf{e}_n^L \alpha_2} \mathbf{m}_n, \alpha_2 \in [0, \frac{\pi}{2}] \quad (8)$$

where $\mathbf{e}_n^U = \mathbf{v}_n \times \mathbf{w}_n$, $\mathbf{e}_n^L = \mathbf{m}_n \times \mathbf{w}_n$, and α_1 and α_2 are the arc segment's angular distance on upper limbs and lower limbs respectively.

Eqs. (7) and (8) can be expanded to the function of $\mathbf{v}_c$, which states that the moving scope of each upper limb or lower limb covers a whole sphere. According to Quaternion's Multiplication, Eq. (7) can be expanded as:

$$\mathbf{r}_{Un} = e^{\mathbf{e}_n^U \alpha_1} \mathbf{v}_n = (\cos \alpha_1 + \mathbf{v}_n \times \mathbf{w}_n \sin \alpha_1) \mathbf{v}_n = \cos \alpha_1 \mathbf{v}_n + \sin \alpha_1 (\mathbf{v}_n \times \mathbf{w}_n) \mathbf{v}_n \quad (9)$$

where $\mathbf{w}_n$ is already known. As we can see, the equation above is the function of $\mathbf{v}_n$. It's very hard to estimate the working scope of $\mathbf{r}_{Un}$ from this equation. However, based on its physical meaning, because the workspace of $\mathbf{v}_n$ is a subset of that of $\mathbf{r}_{Un}$, and $\mathbf{v}_n$'s working scope covers a whole sphere, the workspace of $\mathbf{r}_{Un}$ covers a whole sphere. Hence, though the workspace of the lower limb does not cover a whole sphere, the workspaces of lower limbs and upper limbs are totally overlapped.

2.2. Least interference design of 3-RRR SPM

In the first design stage of the mechanism which we call the theoretical stage, the primary workspace volume of which doesn't take the interference among mechanical parts into account. However, after its manufacturing, the links will become solid entities and collide with each others. For the interference problem on some complex mechanism like SPM, the conventional empirical methods are not enough to work out. Based on all existing researches, a systematic methodology for SPM least interference design is developed as follows. The interference which might occur in SPM can be simplified to the situations between two links.

Suppose that the 3-RRR SPM is located on the sphere surface S^m . The superscript m indicates that it is on the spherical surface m . For controlling variables, only two links L_1^m and L_2^m are considered at a time. L_n^m denotes the collection of the positions of all the points on the link n in the local reference. Before the motion of each link is determined, since two points define a unique line, two endpoints or joints A_1^m and B_1^m on the link 1 need to be designated first. The subscript 1 means that joint belongs to the link 1. The initial simplest shape of one link is the straight line connecting two endpoints. Put the local coordinate system on the link L_1^m , and W_2^m is defined as the collection of all possible moving positions created by all the points on link 2 in the local coordinate

frame. A compact structure conduces to a high stiffness of 3-RRR SPM in application. As a result, the least interference method needs to satisfy two requirements: a) try at best to avoid interference; and b) try at best to arrange the links compactly as long as there is no conflict with the first requirement. The situations where interference happens can be classified into three categories with three corresponding solutions.

2.2.1. Reducing interference on the same surface (The 1st Reduce-Interference Principle, RIP)

In Fig. 2, the interfered part between L_1^m (the black solid line) and W_2^m (the shadowed area) crosses through W_2^m , and two joints are not within it. The situation can be represented in a mathematical way:

$$L_1^m \cap W_2^m \neq \phi \ \& \ \{A_1^m, B_1^m\} \cap W_2^m = \phi. \quad (10)$$

In this situation, the interference can be avoided by modifying the shape of the link L_1^m on the same spherical surface. Generally, the modification can be conducted in two directions P_1 and P_2 . The choice has to be made between these two options by a certain criterion, like no interference with any other links. Otherwise, the shorter one is preferred.

2.2.2. Reducing interference on different surfaces (The 2nd RIP)

See Fig. 3, the second situation occurs when the link L_1^m interferes with W_2^m , and at least one joint of L_1^m is within the workspace W_2^m . Their relation can be represented by:

$$L_1^m \cap W_2^m \not\subset \{A_1^m, B_1^m\} \ \& \ \{A_1^m, B_1^m\} \cap W_2^m \neq \phi. \quad (11)$$

In this situation, there is no room to modify the link shape on the same surface or layer. However, for reducing or avoiding interference, the interfered part of L_1^m can be put on another layer S^{m+1} or S^{m-1} . Accordingly, there are also two ways P_1 and P_2 to reshape the link.

2.2.3. Reducing interference by different joint types (The 3rd RIP)

Similarly, in the third situation, L_1^m interferes with W_2^m on one joint of L_1^m . It also can be represented by the formulas:

$$L_1^m \cap W_2^m \subset \{A_1^m, B_1^m\} \ \& \ \{A_1^m, B_1^m\} \cap W_2^m \neq \phi. \quad (12)$$

According to requirement b), it is better to arrange the links on the same surface than on different surfaces. In this situation, the interference can be avoided or minimized via changing the joint type from Fig. 4a. to Fig. 4b.

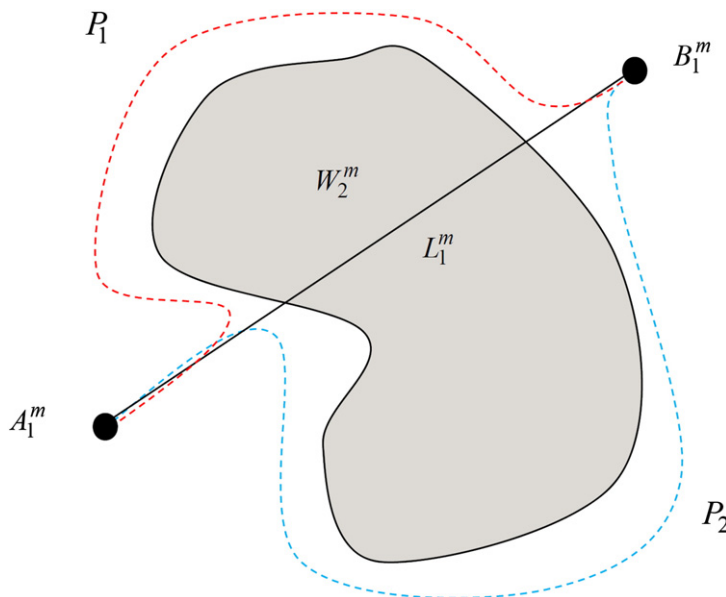


Fig. 2. The 1st Reduce-Interference Problem.

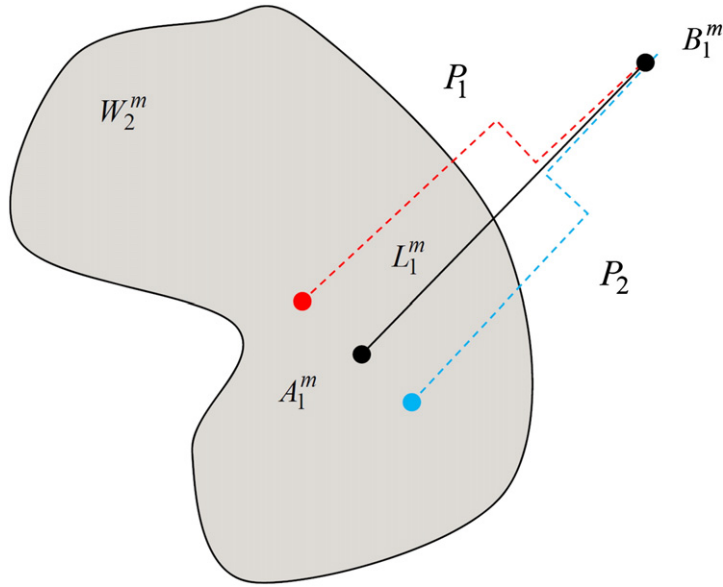


Fig. 3. The 2nd Reduce-Interference Problem.

2.3. The modified structure of 3-RRR SPM according to LIDeM theory

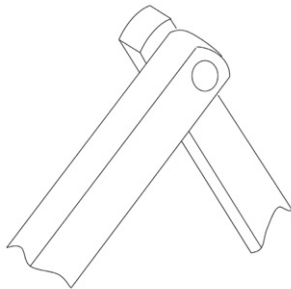
There are two kinds of interferences in the system, one is between the upper limbs and lower limbs, and the other is among upper the limbs or lower limbs. From the above analysis, since the workspaces of upper limbs and lower limbs are totally overlapped, they need to be arranged on different layers (according to the 2nd RIP). By the restriction of topology and isotropy, same level limbs cannot be put on different layers. However, the 2nd RIP can still be used to reduce interference partly through arranging a part of the link onto another layer. After modification, a pair of limbs is shown in Fig. 5. β_1 and β_2 are designated as the ratios of the modified section length (the yellow arcs) to the total one on upper and lower limbs respectively.

Consequently, the whole structure of the 3-RRR SPM can be seen in Fig. 6. For optimum workspace, β_1 and β_2 remain to be determined.

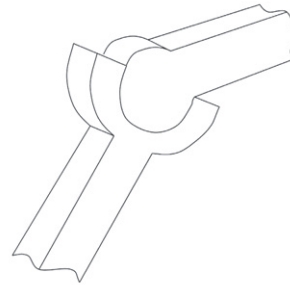
3. Optimum workspace synthesis

The intersection point between every two upper limbs is derived:

$$\begin{cases} \mathbf{s}_{Un} = e^{\mathbf{e}_n^U \theta_n^{Un}} \mathbf{v}_n \\ \mathbf{s}_{Un} = e^{\mathbf{e}_n^U \theta_{n+1}^{Un}} \mathbf{v}_{n+1} \end{cases} \quad (13)$$



a) the revolute joint between two surfaces



b) the revolute joint on one surface

Fig. 4. The 3rd Reduce-Interference Problem, a. the revolute joint between two surfaces, b. the revolute joint on one surface.

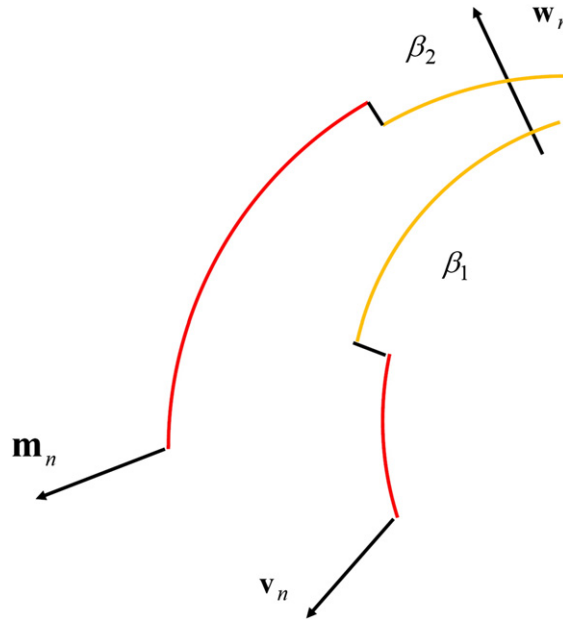


Fig. 5. A pair of limbs after modification according to the 2nd RIP.

when n is equal to 3, $n + 1 = 1$. Likewise, for the lower limbs:

$$\begin{cases} \mathbf{s}_{Ln} = e^{\mathbf{e}_n^L \sigma_n^{Ln}} \mathbf{m}_n \\ \mathbf{s}_{Ln} = e^{\mathbf{e}_n^L \sigma_{n+1}^{Ln}} \mathbf{m}_{n+1} \end{cases} \quad (14)$$

where $\sigma_n^{Un} \in [0, \pi/2]$ represents the angle between the intersection point and the revolute vector $\mathbf{w}_n$ on the upper limbs, and σ_n^{Ln} is for the lower limbs. Fig. 7 indicates that interference occurs under situations (a) and (b), but not under (c). For upper limbs, the condition without interference is given as:

$$C_1 : \left(\sigma_n^{Un} - \beta_1 \frac{\pi}{2} \right) \left(\sigma_{n+1}^{Un} - \beta_1 \frac{\pi}{2} \right) < 0 \quad (15)$$

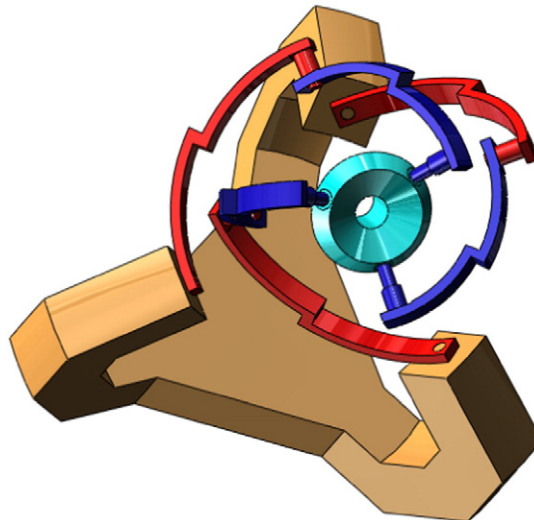


Fig. 6. The modified structure of 3-RRR SPM.

and likewise for the lower limbs:

$$C_2 : \left(\sigma_n^{Ln} - \beta_2 \frac{\pi}{2} \right) \left(\sigma_{n+1}^{Ln} - \beta_2 \frac{\pi}{2} \right) < 0. \quad (16)$$

According to the method for measuring the workspace developed by Gosselin and Angeles [14], the total workspace without interference is also a volume, and can be calculated from the equation:

$$WS(\beta_1, \beta_2) = \int_{C_1, C_2} \eta \Omega(\phi_1, \phi_2, \beta_1, \beta_2) dS \quad (17)$$

where Ω is the size of the interval of ω ; dS is the infinitesimal of the sphere whose total surface area is equal to 1. For the isotropic mechanism design, let the weight coefficient η be equal to $1/S = 1$, whereas it can be a variable when it comes to the non-isotropic design. The expanded expression of Eq. (17) is given as:

$$WS(\beta_1, \beta_2) = R^2 \iint_{C_1, C_2} \Omega(\phi_1, \phi_2, \beta_1, \beta_2) \sin \phi_2 d\phi_1 d\phi_2 \quad (18)$$

where R represents the radius of the sphere S and is equal to $0.5/\pi^{0.5}$.

After the numerical computation for Eq. (18), the function of the total workspace volume with respect to variable β_1 and β_2 is shown in Fig. 8. Evidently, when $\beta_1 = \beta_2 = 0.64$, the total workspace reaches the maximum value 5.4946 (rad).

For intuitively displaying how much the method contributes for the total workspace, the “tilt & torsion angles” method is adopted. Because spherical surface and planar surface are locally homeomorphous, it is more precise to illustrate these angles in a spherical coordinate system rather than in a cylindrical one. Every point on the volume surface can be located by the coordinates (Ω, ϕ_1, ϕ_2) . Fig. 9 shows the workspace volume of 3-RRR SPM with the optimum structure, and Fig. 10 indicates the result of the Agile Eye [7] computed by the authors. Apparently, the workspace of 3-RRR SPM with the optimum structure manifests a larger volume.

4. Singularities analysis and interfered areas

Singularities will cause mechanisms to gain or lose some DOFs, and even worse to be damaged. It is vital to locate all the singular points in order to avoid them by motion planning. In the 3-RRR SPM, all the singular points can be defined by a Jacobian Matrix [15]:

$$\mathbf{J}_n = \frac{(\mathbf{w}_n \times \mathbf{v}_n)^T}{(\mathbf{m}_n \times \mathbf{w}_n) \cdot \mathbf{v}_n}. \quad (19)$$

According to this matrix, all the singular points appear under two conditions. One is when $\det(\mathbf{J})$ is equal to zero. According to the necessary and sufficient condition for determination equal to zero, this kind of singularity exists only when any two rows of $\mathbf{J}$ should be linearly dependent. However, this condition will never be met for the particular 3-RRR SPM investigated in this paper. For example, postulate that $\mathbf{w}_1 \times \mathbf{v}_1$ and $\mathbf{w}_2 \times \mathbf{v}_2$ are linearly dependent, which leads to $(\mathbf{w}_1 \times \mathbf{v}_1) \times (\mathbf{w}_2 \times \mathbf{v}_2) = 0$. It can be expanded as:

$$(\mathbf{w}_1 \cdot (\mathbf{w}_2 \times \mathbf{v}_2)) \mathbf{v}_1 - (\mathbf{v}_1 \cdot (\mathbf{w}_2 \times \mathbf{v}_2)) \mathbf{w}_1 = 0. \quad (20)$$

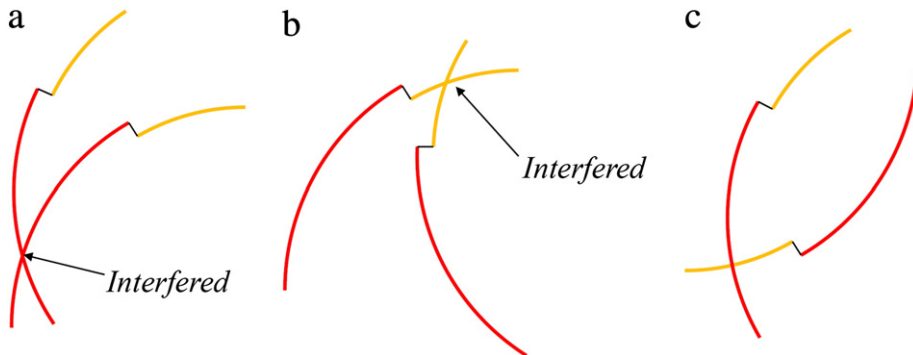


Fig. 7. Three situations with or without interference.

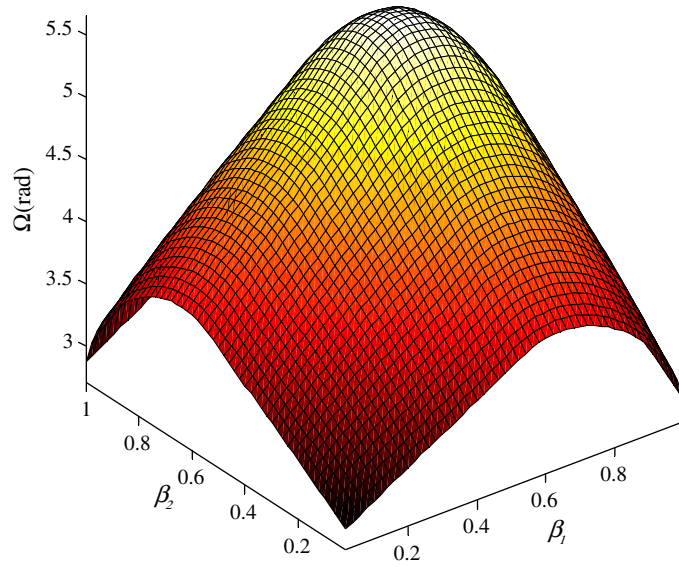


Fig. 8. Total workspace function with variables β_1 and β_2 .

$\mathbf{w}_1$ and $\mathbf{v}_1$ are orthogonal to each other, or in other words, they are linearly independent. Eq. (20) is proven to be false. Consequently, this kind of singularity doesn't exist. The other condition of singularity is when $\det(\mathbf{J}) = \infty$ and $(\mathbf{m}_n \times \mathbf{w}_n) \cdot \mathbf{v}_n = 0$, which indicates that $\mathbf{m}_n$, $\mathbf{w}_n$ and $\mathbf{v}_n$ are coplanar. With this condition, the singularities can be classified into two categories: the first is totally folded, and the second is totally unfolded. Similar with the plots in Figs. 9 and 10, the singularity loci can be visualized by a set of discretized singular points (ω, ϕ_1, ϕ_2) in a spherical coordinate system (Fig. 11). When the mechanism is at the former singular positions, the end-effector cannot respond to the inputs, which is considered as the loss of one DOF. For the latter singularity, it gains one extra DOF. When the DOF suddenly increases from 3 to 4, the system becomes unstable, and its motion cannot be determined.

Generally, the primary workspace of a certain mechanism is larger than its reachable workspace. Because of the influence of interference to the workspace, to optimize the workspace of 3-RRR SPM is actually to minimize the interference of it, and to get

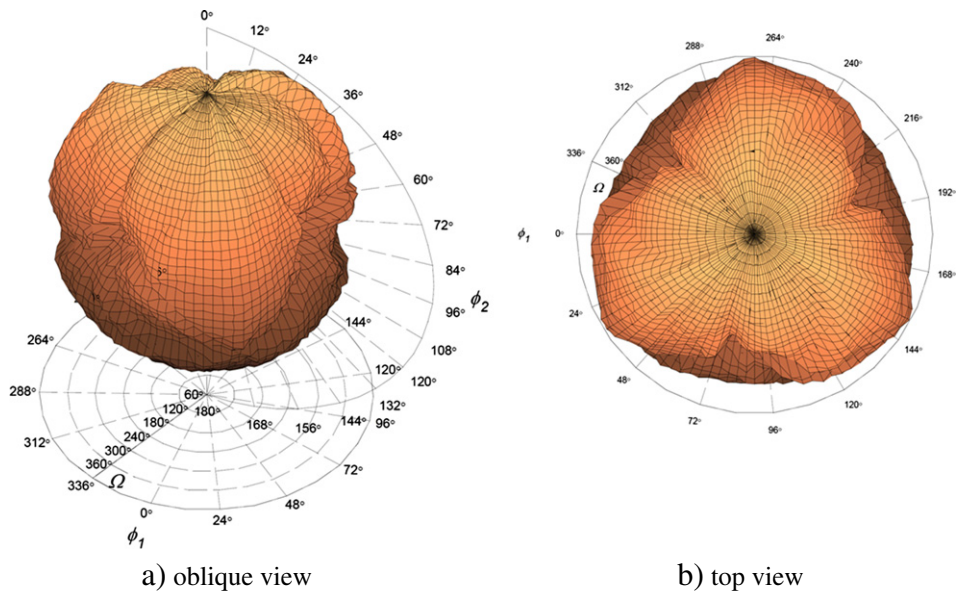


Fig. 9. Total workspace of 3-RRR SPM with the optimum structure, a. oblique view, b. top view.

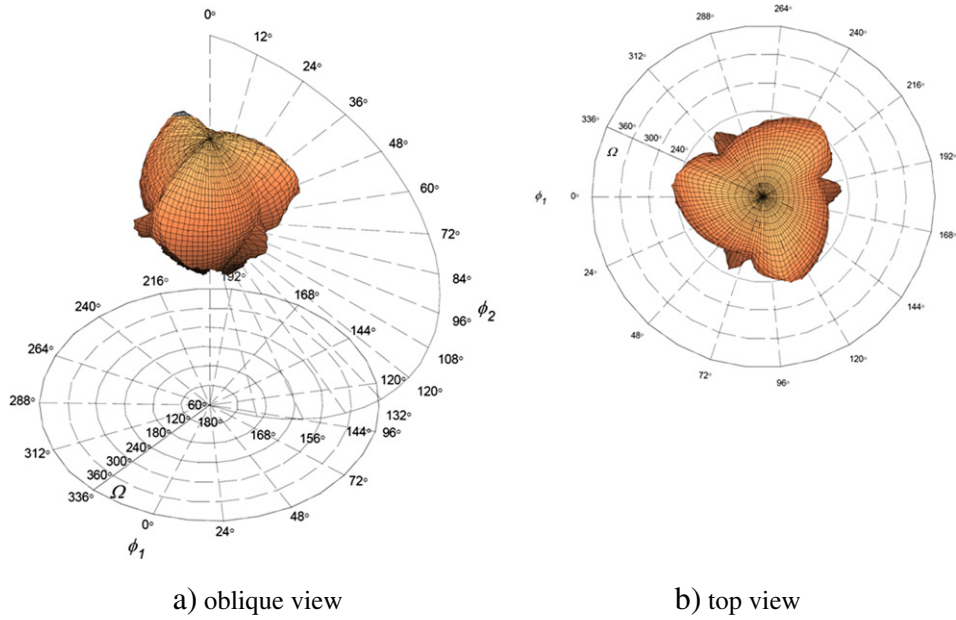


Fig. 10. Total workspace of the Agile Eye, a. oblique view, b. top view.

the maximum free-interference workspace volume. Hence, the optimum workspace in this paper is the minimum-interference reachable workspace.

In addition, although the system has the optimum workspace, the interference in some area cannot be avoided. Even if with minimum interference among the links, there would be some orientations which the end-effector can never reach. These unreachable orientations are actually the holes and voids shown in Fig. 12. All the unreachable orientations are not taken into account while calculating the volume of the optimum workspace. In fact, the optimum workspace is equal to the primary workspace minus these holes and voids.

Because inertia exists, for avoiding collisions, the end-effector needs to be slowed down before it enters these areas. Through numerical simulation, the distributions of interference area are worked out. In Fig. 12a–d, the red areas are the free positions, while the white holes and voids are with interference with respect to a certain ω .

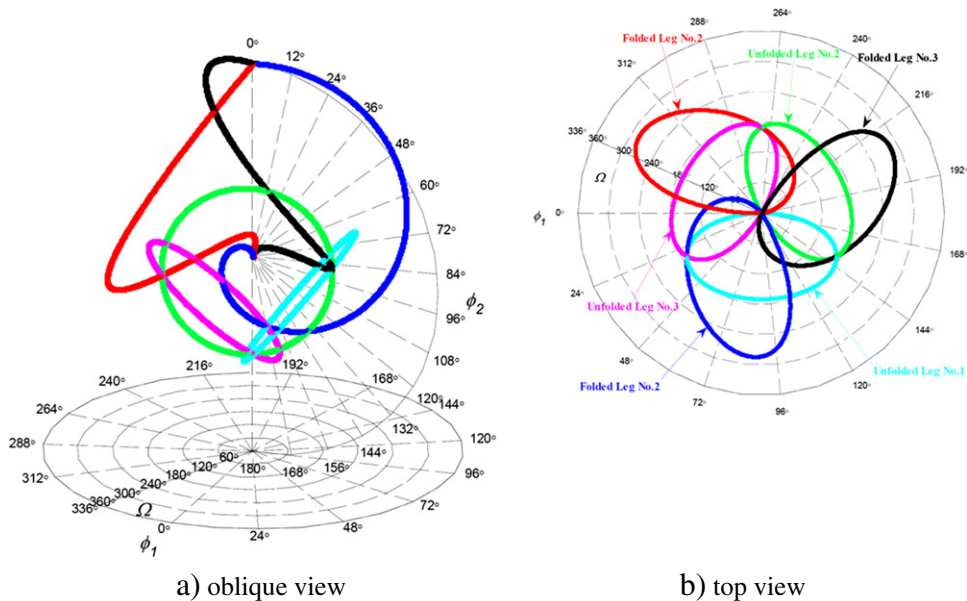


Fig. 11. Singularity loci, a. oblique view, b. top view.

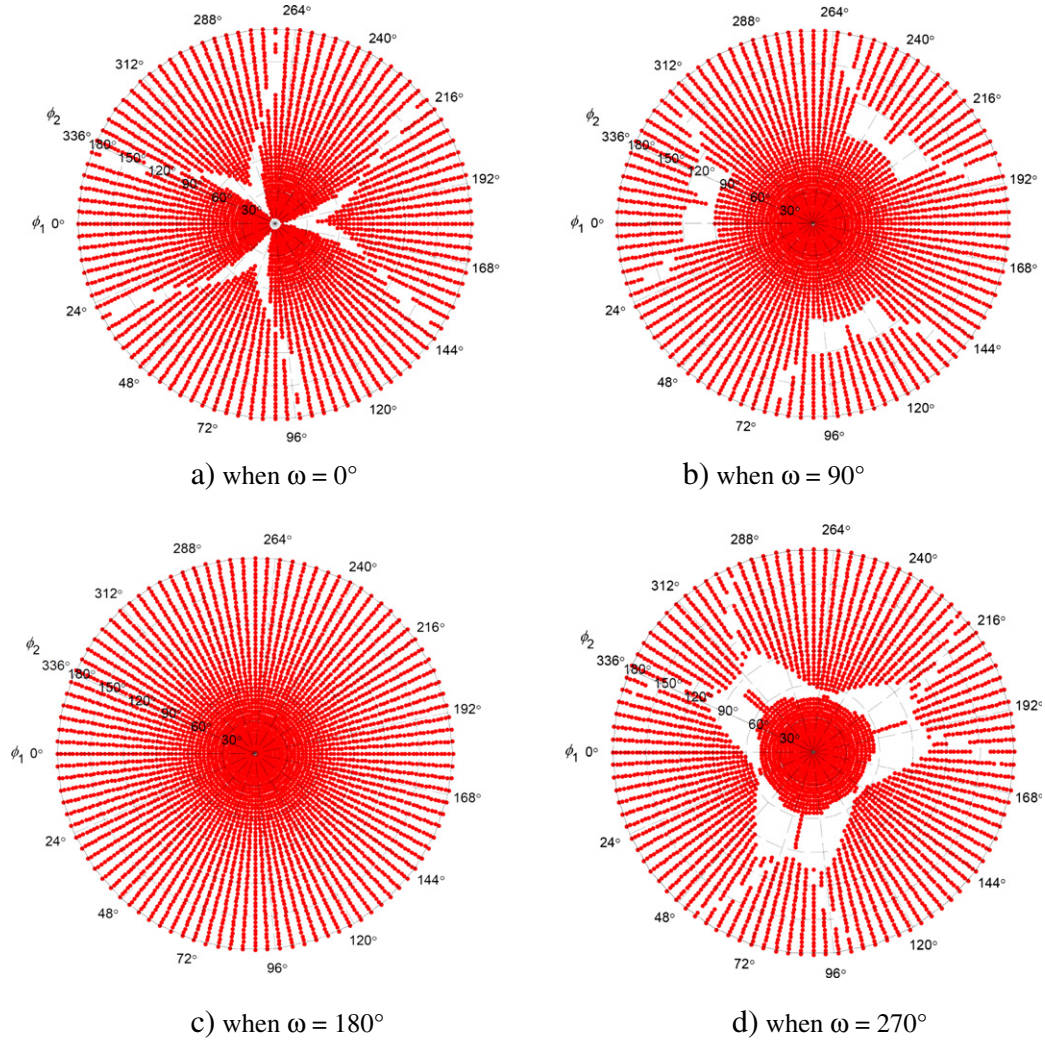


Fig. 12. The distribution of interference areas, a. when $\omega = 0^\circ$, b. when $\omega = 90^\circ$, c. when $\omega = 180^\circ$, d. when $\omega = 270^\circ$.

5. Conclusions

- (1) Based on the position analysis of 3-RRR SPM and all existing researches on avoiding interference, the LIDeM theory is proposed. According to this methodology, a new 3-RRR SPM with optimum workspace is developed.
- (2) The conditions whether interference happens are given by the mathematical modeling of upper limbs and lower limbs. Using these conditions, the total workspace volume function with respect to the link ratio β_1 and β_2 is plotted. It indicates that when $\beta_1 = \beta_2 = 0.64$, the workspace volume reaches the maximum value 5.4946 (rad).
- (3) Through numerical simulation, the workspace of the optimum structure with respect to tilt and torsion angles is also plotted. It shows that the optimum mechanism has a larger working range evidently. The singularity loci and interference distributions are illustrated by the mapping in a spherical coordinate system.

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